BENJAMIN ANDERSON

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Electronic Engineer

PROJECTS

Cube
A handheld device comprising of 6 OLED screens for each face of a cube. This project required design of circuits for battery management and voltage regulation, 3D modelling of the frame, as well as efficient C code to run 6 screens and a 6 DOF sensor off a single ESP32.
`KiCAD / FreeCAD / Git / Blender`

Orthodox
A unique ortholinear keyboard. This project required rapid prototyping of custom keycaps using parametric designs, PCB coils around keyswitches to measure analog keypresses with an LDC chip, as well as a custom driver had to be written to allow for seamless integration of the keyboard with computers
`KiCAD / FreeCAD / Git / OpenSCAD`

EXPERIENCE

RF Test Engineer — **Curvalux** 2021-2022
- Used LaTeX to produce clean, formatted reports of data/graphs generated with matplotlib.
- Professionally collaborated and documented repositories using gitlab, jira, confluence and slack.
- Developed python libraries to control rotators, switches, keysight VNAs and beamforming ICs for automating radiation pattern measurements in a custom anechoic chamber.
- Translated MATLAB algorithms for phase tapering/beam steering a phased array antenna into working code on early prototype hardware.

Research & Development Engineer — **Huawei** 2019-2020
- Analysing reliability of devices with PAM4 / NRZ inspection, shear testing, electroluminescence inspection.
- Built a standalone program with PyQt to automatically produce failure analysis reports from data gathered by lab technicians, saving countless engineering hours on a repetitive manual task.
- Developed a python library to communicate with a TEC controller over the serial protocol.
- Presented weekly meetings to inform senior engineers of device failures and reliability.

Hardware Engineer — **Avalon** 2018-2019
- Tasked with designing and soldering the power management PCB for an underwater ROV
- Had meetings with suppliers to discuss partnership/sponserships for the competition.
- Regularly liaised with engineers from different disciplines to check compatibility.
- Managed my time effectively to balance commitments to University and Avalon.

Design Project — **Global Engineering Challenge** 2018-2019
- Tasked with diagnosing diseases in Kibera, exploratory research led to designing mobile retina scanners.
- Used TensorFlow and an NHS dataset to train a model to determine diseases based on symptoms visible in retinas.
- Pitched the proof of concept in front of a crowd of 30+ people, receiving questions for 10 minutes.

REFERENCES

Jason Hanson — **Princial RF Engineer**
jason.hanson@curvalux.co.uk
+44 (0)7799 036 362

Xiaoli Chu — **Ph.D Wireless Communications**
x.chu@sheffield.ac.uk
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Lu Zhang — **Senior Reliability Engineer**
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EDUCATION

Bachelor of Engineering — **University of Sheffield** 2017-2021
Electronic Engineering: 1st

A-Levels — **Dunraven** 2015-2017

Mathematics	B
Physics	B
Computing	B

GCSEs (11 Qualifications Passed) — **Harris Academy** 2010-2015

Mathematics	A
Science	A
English	B